Import Data using Transform Maps

(Spreadsheet to ServiceNow)

Employee Data Import using Transform Maps (Spreadsheet) - ServiceNow

Project Report	
Team ID	
Team Members	
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1. INTRODUCTION

1.1 Project Overview

Organisations often receive employee-related data in spreadsheets (Excel or CSV). Entering this data record by record into ServiceNow is slow and error-prone, and the records are not automatically linked to employees. This project uses ServiceNow **Import Sets** and **Transform Maps** to load spreadsheet data into a staging table and transform it into a target table, linking each record to an employee (Employee ID mapped to the sys_user reference field) and pulling details such as department into the record for easier reporting.

The imported data is then protected with **Access Control Lists (ACLs)**, and the access rules are tested using the **Impersonate User** feature.

1.2 Purpose

The purpose of the project is to:

- Import bulk spreadsheet data quickly and accurately without manual entry.
- Link every record to an employee and bring in department details for reporting.
- Prevent duplicate records using Coalesce.
- Implement ACLs and verify them with Impersonate User, giving a solid foundation for managing and securing data in ServiceNow.

2. IDEATION PHASE

2.1 Problem Statement

Problem statement: Linking each record to an employee and pulling some employee details (like department) into the record for easier reporting.

Objective: To implement an Access Control List (ACL) by using Impersonate User. This provides a solid foundation for managing and securing data in ServiceNow using key platform features.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A ServiceNow administrator / HR staff who receives employee data in spreadsheets	Import spreadsheet data using Transform Maps and link each record to an employee with department details	Manual entry is slow and department details are missing or inconsistent	Data is not mapped to the Employee (sys_user) reference field and has no coalesce rule	Frustrated and overwh elmed
PS-2	A system administrator / security owner	Control who can view or edit employee-linked records	Any user may see or change sensitive fields	No ACLs are defined and access has not been tested using Impersonate User	Worried about data safety

2.2 Empathy Map Canvas

The empathy map captures the viewpoint of the ServiceNow administrator who imports employee data from spreadsheets.

Quadrant	Observations
Think & Feel	Will my data map correctly? Is department filled in? Will duplicates be created? Is the data secure?
Hear	Reports need department information; do not forget to set Coalesce; check Transform History.
See	Spreadsheets with hundreds of rows; blank or mismatched columns; ignored rows in the log; records not linked to an employee.
Say & Do	Create the Data Source and attach the file; map Employee ID to sys_user; run the transform and check results; test ACLs with Impersonate User; fix errors and re-import.
Pain	Manual data entry; duplicate or ignored records; missing department details.
Gain	Fast bulk import; accurate employee linking; secure access using ACLs.

2.3 Brainstorming

The team brainstormed ideas, grouped them into clusters and then prioritised them on an importance versus feasibility grid.

Cluster	Ideas
A: Import setup	Create Data Source and attach file; clean headers and blank rows; check Transform History.
B: Employee linking	Map Employee ID to the reference field; set Coalesce on Employee ID; auto-pull department.
C: Security (ACL)	Create ACL on the Department field; role-based access; test using Impersonate User.
D: Automation & Reporting	Scheduled import; onBefore script to skip invalid rows; department-wise report.

Top priority ideas selected: create a Data Source and load the spreadsheet, set Coalesce on Employee ID, map Employee ID to a reference field on sys_user, and secure the data with ACLs verified by Impersonate User.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey of the administrator is described in five stages.

Stage	What happens	Pain point	Opportunity
Entice	Receives the employee spreadsheet from HR and decides to use Import Set and Transform Map	Spreadsheets arrive in different formats	Standardise the spreadsheet template
Enter	Cleans the file, creates the Data Source, attaches the file and loads records	Wrong sheet or header setting loads no rows	Validate files before loading
Engage	Creates the Transform Map, maps fields (Employee ID to sys_user), sets Coalesce and runs the transform	Reference field not matching leaves values empty	Use a script to auto-fill department
Exit	Reviews Transform History, verifies records, creates ACLs and tests with Impersonate User	Ignored or skipped rows are hard to trace	Alert on failed rows; test several roles
Extend	Uses employee-linked records in reports, schedules imports and reuses the Transform Map	Changed data needs repeated imports	Build department dashboards; schedule imports

3.2 Solution Requirement

Functional requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	Spreadsheet Data Preparation	Prepare the file with a header row; clean blank rows and inconsistent formats
FR-2	Data Source and Data Loading	Create a Data Source and attach the file; load records into the Import Set table
FR-3	Transform Map Configuration	Create the Transform Map; map fields; map Employee ID to sys_user reference; set Coalesce
FR-4	Employee Details Population	Pull department and other employee details into each record
FR-5	Transform Execution and Verification	Run the Transform; review Transform History and Import Log
FR-6	Access Control (ACL)	Create ACLs on the target table and fields; test using Impersonate User
FR-7	Reporting	Create reports and list views grouped by department

Non-functional requirements

NFR No.	Requirement	Description
NFR-1	Usability	Uses standard ServiceNow modules and a simple spreadsheet template; no coding needed
NFR-2	Security	Role-based ACLs verified with Impersonate User; only authorised users can import
NFR-3	Reliability	Coalesce prevents duplicates; Transform History makes errors traceable

NFR No.	Requirement	Description
NFR-4	Performance	Bulk rows loaded and transformed in one run; large files can be split into batches
NFR-5	Availability	Runs on the ServiceNow cloud; imports can be scheduled off-peak
NFR-6	Scalability	The Transform Map is reusable for new files and larger data volumes

3.3 Data Flow Diagram

The Level 0 data flow diagram has three external entities (HR data owner, Administrator and End Users), three processes and four data stores.

Element	Name	Description
Entity	HR (Data Owner)	Provides the employee spreadsheet to the Administrator
Entity	Administrator	Configures the Data Source, Transform Map and ACLs, and tests with Impersonate User
Entity	End Users	Receive only the records and fields they are permitted to see
Process 1.0	Load Data	Loads the attached file into the staging table
Process 2.0	Transform Data	Applies the Transform Map to produce employee-linked records
Process 3.0	Apply and Test ACLs	Secures the target table and verifies access
Data store D1	Import Set Table (u_)	Holds the raw staged rows
Data store D2	Employee Table (sys_user)	Source of employee details such as department
Data store D3	Target Table	Stores the final employee-linked records
Data store D4	Transform History	Stores counts of inserted, updated and ignored rows

3.4 Technology Stack

Component	Technology
User Interface	ServiceNow Next Experience UI (HTML, CSS, JavaScript)
Application Logic	Data Source, Import Set, Transform Map and Field Maps; field map scripts (JavaScript, GlideRecord)
Database	ServiceNow tables: staging (u_) table, target table, Employee (sys_user)
File Storage	ServiceNow Attachments (sys_attachment)
External APIs / ML	Not required (REST Import Set API and LDAP integration are optional); no machine learning model
Infrastructure	ServiceNow Cloud (Personal Developer Instance); Chrome / Edge and Microsoft Excel locally
Security	ACLs, roles, Impersonate User, HTTPS / TLS

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem-solution fit confirms that importing spreadsheet data with Transform Maps solves the administrator's real problem.

Canvas element	Summary
Customer segment	ServiceNow administrators and HR data owners who receive employee data as Excel / CSV
Customer constraints	Limited time, little coding skill, inconsistent spreadsheets, need for secure access
Available solutions	Manual entry (slow), copy-paste (no employee link), custom scripts (complex)
Jobs-to-be-done / problems	Import bulk data, link to employees, pull department, secure with ACLs
Problem root cause	No field mapping to sys_user, no Coalesce rule, no ACLs defined or tested
Triggers	New employee sheet from HR; request for department-wise reports; audit questions
Emotions	Before: frustrated and worried. After: confident and in control
Our solution	Import Sets and Transform Maps with Coalesce, department script and ACLs tested using Impersonate User
Channels	Online: ServiceNow UI, email, documentation. Offline: Excel files and team discussions

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement	Manual, error-prone entry of spreadsheet data; records not linked to employees; no access control
2.	Idea / Solution description	Import with a Data Source, staging table and Transform Map; link Employee ID, use Coalesce, pull department; secure with ACLs tested using Impersonate User
3.	Novelty / Uniqueness	Combines bulk import, employee linking and ACL testing in one low-code workflow using standard ServiceNow features
4.	Social Impact / Customer Satisfaction	Saves hours of manual work, reduces errors and protects employee information
5.	Business Model	Internal productivity solution on an existing ServiceNow subscription; reusable as a template or service
6.	Scalability	Reusable maps, batch loading, scheduled imports, more tables and roles

4.3 Solution Architecture

The architecture and data flow of the solution is as follows:

1. The Administrator prepares the employee spreadsheet (Excel / CSV) and attaches it to a Data Source.
2. The records are loaded into the Import Set (staging) table.
3. The Transform Map reads the staged rows.
4. Field map scripts transform values where needed.
5. Employee details (department) are looked up from the Employee (sys_user) table.
6. Linked records are written to the Target table and the run is logged in Transform History.
7. ACLs secure the Target table and are tested with Impersonate User; permitted data is shown to end users and in reports.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was planned in four sprints of 6 days each with a velocity of 20 story points per sprint. Average velocity = 20 / 6 = 3.33 story points per day.

Sprint	Stories	Focus	Story Points	Start Date	End Date
Sprint-1	USN-1 to USN-4	Data Source, loading, Transform Map, field mapping	20	05 Oct 2026	10 Oct 2026
Sprint-2	USN-5 to USN-7	Coalesce, department population, verification	20	12 Oct 2026	17 Oct 2026
Sprint-3	USN-8 to USN-11	ACLs, Impersonate User testing, reporting, access control	20	19 Oct 2026	24 Oct 2026
Sprint-4	USN-12, USN-13	Scheduled import, testing and documentation	20	26 Oct 2026	31 Oct 2026

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The following test cases verify the functionality and performance of the import. Record the actual result and status after running each test on your ServiceNow instance.

ID	Test case	Input	Expected result	Actual result / Status
T-1	Load spreadsheet into staging table	Sample file with 3 to 10 employee rows	All rows appear in the Import Set table	
T-2	Run Transform Map	Import Set from T-1	Records inserted in the target table; counts shown in Transform History	
T-3	Employee link and department	Valid Employee IDs	Each record links to the employee and shows the department	
T-4	Coalesce (re-import)	Same file loaded again	Existing records updated; no duplicates created	
T-5	Invalid data handling	Blank or unknown Employee ID	Row is ignored or flagged; error visible in the Import Log	
T-6	ACL access	Impersonate an unauthorised user	Restricted fields and records are hidden or read-only	
T-7	ACL access (authorised)	Impersonate an authorised role	User sees the permitted records and fields	
T-8	Larger file performance	File with a few hundred rows	Import completes without errors; note the time taken	

7. RESULTS

7.1 Output Screenshots

Insert the screenshots from your ServiceNow instance in the spaces below.

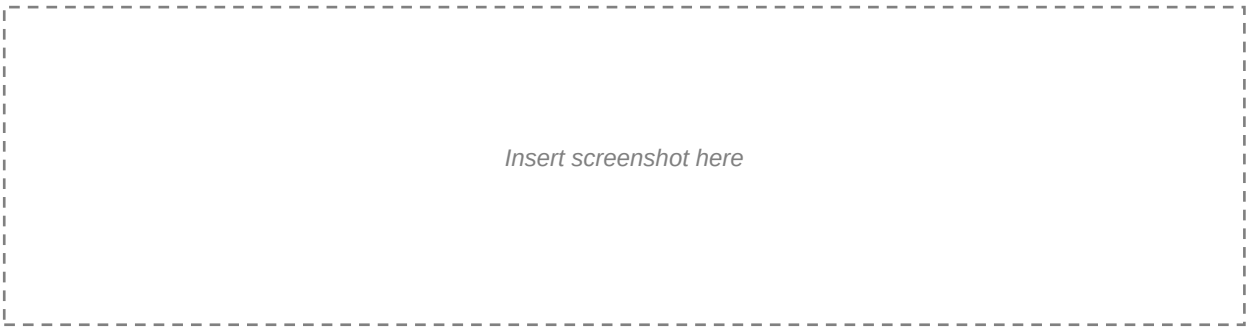

Figure 1: Data Source with the spreadsheet attached

Figure 2: Import Set table showing the loaded rows

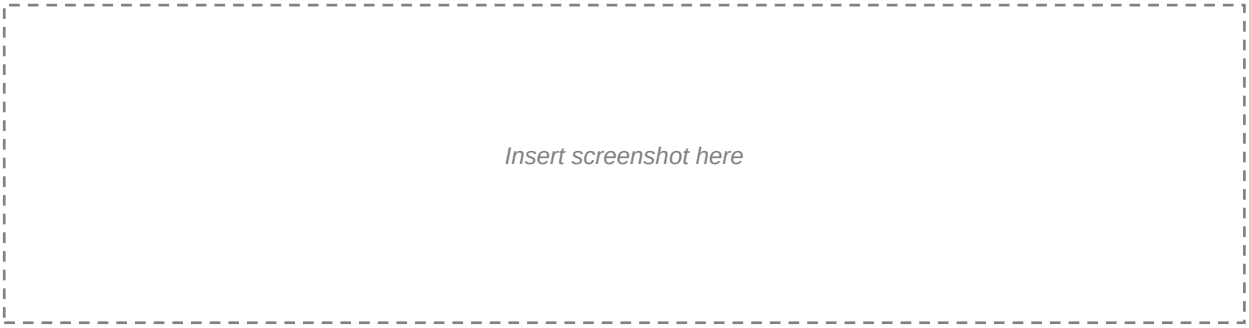

Figure 3: Transform Map and Field Maps (Employee ID to sys_user, Coalesce)

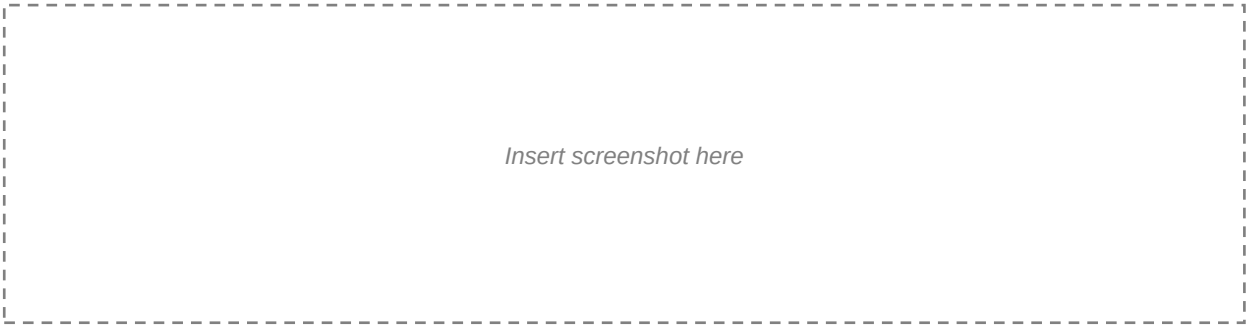

Figure 4: Transform History showing inserted / updated counts

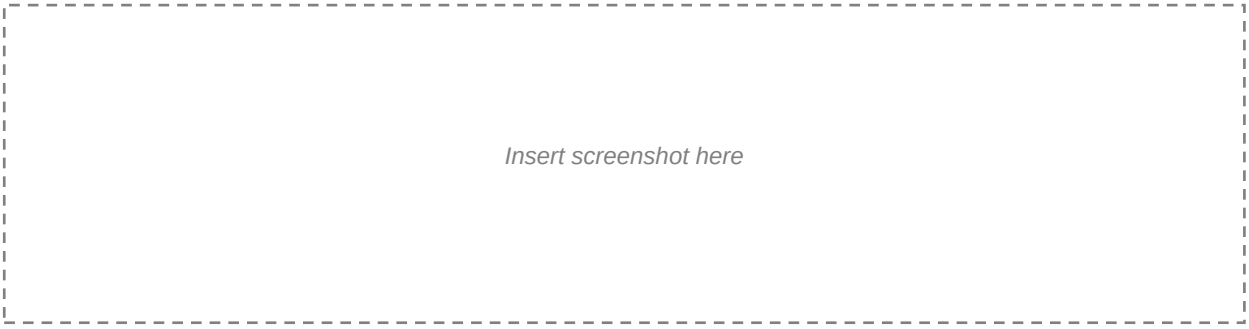

Figure 5: Target table with employee link and department populated

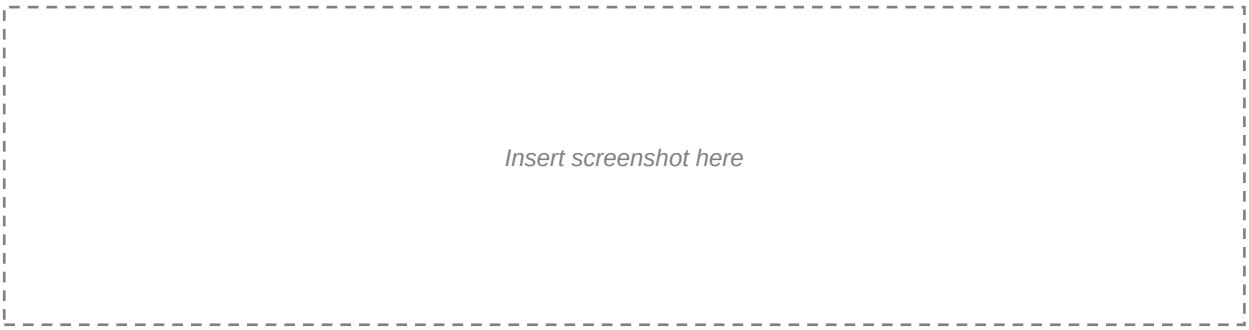

Figure 6: ACL record created for the target table / field

Figure 7: Impersonate User test showing restricted access

8. ADVANTAGES & DISADVANTAGES

Advantages

- Saves time compared with manual data entry.
- Reduces data-entry errors and enforces consistent mapping.
- Links each record to an employee and brings in department details automatically.
- Coalesce prevents duplicate records and allows updates.
- ACLs protect sensitive data and can be verified quickly with Impersonate User.
- Reusable and schedulable; uses only standard ServiceNow features.

Disadvantages

- The spreadsheet must be clean and consistently formatted before import.
- Mapping and scripting require some ServiceNow knowledge.
- Reference fields can stay empty if values do not exactly match existing records.
- Very large files may need to be split into batches.
- Incorrect ACLs can block valid users, so they need careful testing.

9. CONCLUSION

Transform Maps provide a reliable and controlled way to move spreadsheet data into ServiceNow. By preparing clean data, defining a Data Source, mapping fields, linking Employee ID to the Employee table and using Coalesce, administrators can import and update records quickly and accurately. Pulling department details into the record supports easier reporting, and ACLs tested with Impersonate User secure the data. The project meets its objective of providing a solid foundation for managing and securing data in ServiceNow.

10. FUTURE SCOPE

- Schedule imports to refresh employee data automatically.
- Integrate directly with an HR system or LDAP instead of spreadsheets.
- Add data validation scripts and error notifications for failed rows.
- Extend the approach to other tables such as assets, incidents and departments.
- Build dashboards for department-wise analytics.
- Apply more detailed role-based ACLs and audit reports.

11. APPENDIX

Source Code (if any)

Sample field map script to store the e-mail address in lower case:

```
answer = (function transformEntry(source) {
  return source.u_email.toString().toLowerCase();
})(source);
```

Sample dataset columns: First Name, Last Name, Employee ID, Email, Department.

Dataset Link

Dataset link: _____

GitHub & Project Demo Link

GitHub link: _____

Project demo link: _____